❤️■ Heart Disease Prediction

An end-to-end machine learning pipeline using Random Forest to predict the likelihood of heart disease from clinical patient data, with an interactive Streamlit deployment.

Python 3.x	scikit-learn	Random Forest	Streamlit App

1. Project Overview

This project builds a full clinical prediction pipeline for heart disease using the UCI Heart Disease dataset. It covers exploratory analysis, outlier handling, categorical encoding, feature correlation analysis, and Random Forest classification — deployed via a Streamlit web app for real-time patient risk scoring.

01	02	03	04	05	06
Data Loading	EDA & Profiling	Cleaning	Encoding	RF Model	Streamlit App

2. Dataset Summary

918	11	2	~55%
PATIENTS	FEATURES	TARGET CLASSES	POSITIVE RATE

Numerical features
Age, RestingBP, Cholesterol, MaxHR, Oldpeak — continuous clinical measurements subject to outlier analysis via Z-score.

Categorical features
Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope, FastingBS — one-hot encoded via get_dummies(drop_first=True).

Target variable
HeartDisease: 0 = no heart disease, 1 = heart disease present. Mildly imbalanced (~45% vs ~55%).

Data quality actions
Z-score outlier removal (threshold=3), zero RestingBP rows dropped, Cholesterol zeros imputed with group median.

3. Data Cleaning

Step 1 — Outlier removal via Z-score

Z-scores computed for Age, RestingBP, Cholesterol, FastingBS, MaxHR, Oldpeak. Rows where any feature exceeded z=3 (~3 standard deviations) were removed. This conservative threshold preserves most data while eliminating extreme values that could bias the model.

Step 2 — RestingBP zero removal

A resting blood pressure of 0 mmHg is physiologically impossible and represents missing or erroneous data. These rows were dropped entirely rather than imputed to avoid introducing false signal.

Step 3 — Cholesterol zero imputation

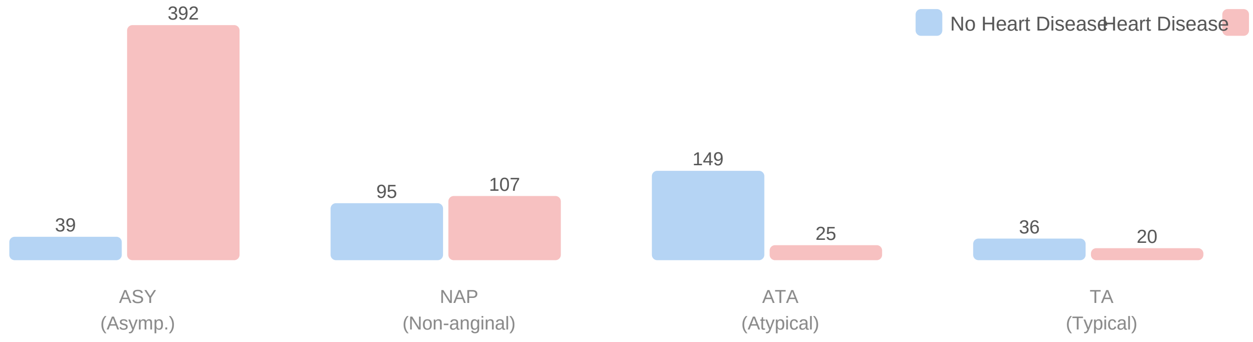
Zero Cholesterol values (a missing-data proxy) were imputed with the group-specific median — separately for patients with and without heart disease. This preserves the clinical relationship between cholesterol and diagnosis rather than using a single global median that would dilute the signal.

```
# Z-score outlier removal z_scores = np.abs(zscore(df_subset)) data = heart_df[~(z_scores >
3).any(axis=1)] # Group-median cholesterol imputation mask = df_clean["HeartDisease"] == 0
df_clean.loc[mask,"Cholesterol"] = chol_no_hd.replace(0, chol_no_hd.median())
df_clean.loc[~mask,"Cholesterol"] = chol_hd.replace(0, chol_hd.median())
```

4. Exploratory Data Analysis

Chest pain type vs heart disease

Count plots were generated for all 7 categorical columns. ASY (asymptomatic) chest pain is by far the most dangerous type — 392 of 431 ASY patients (91%) have heart disease, despite feeling no chest pain. This counterintuitive finding is clinically significant: asymptomatic patients should not be ruled out based on absence of pain alone. Conversely, ATA (atypical angina) patients mostly do not have heart disease.



Chest pain type distribution (pie)

ASY dominates the dataset at 54%, meaning over half the patients present with no chest pain. NAP (non-anginal) accounts for 22%, ATA for 19%, and classic TA (typical angina) is rare at only 5% — challenging the assumption that typical anginal symptoms drive most diagnoses.

Sex vs heart disease

Males represent ~79% of the dataset. Of ~725 males, roughly 62% are positive for heart disease. Females show a much lower positive rate (~26%). This reflects both genuine clinical prevalence differences and potential dataset collection bias. The model should not be over-generalised to female populations without further validation data.

5. Correlation Heatmap

Absolute correlations were computed after one-hot encoding all categorical variables. Two heatmaps were plotted: the full matrix and a filtered view (correlations > 0.3). The rocket_r colormap highlights the strongest associations in dark red.

	HeartDis	ST_Up	ST_Flat	ExAngina	Oldpeak	MaxHR	Age	Cholest	RestBP	FastBS
HeartDis	1.00	0.59	0.52	0.49	0.40	0.38	0.21	0.09	0.07	0.22
ST_Up	0.59	1.00	0.72	0.41	0.43	0.37	0.14	0.07	0.05	0.13
ST_Flat	0.52	0.72	1.00	0.36	0.51	0.34	0.18	0.06	0.07	0.16
ExAngina	0.49	0.41	0.36	1.00	0.36	0.39	0.10	0.04	0.06	0.12
Oldpeak	0.40	0.43	0.51	0.36	1.00	0.35	0.18	0.10	0.09	0.11
MaxHR	0.38	0.37	0.34	0.39	0.35	1.00	0.35	0.09	0.08	0.07
Age	0.21	0.14	0.18	0.10	0.18	0.35	1.00	0.06	0.12	0.09
Cholest	0.09	0.07	0.06	0.04	0.10	0.09	0.06	1.00	0.05	0.04
RestBP	0.07	0.05	0.07	0.06	0.09	0.08	0.12	0.05	1.00	0.03
FastBS	0.22	0.13	0.16	0.12	0.11	0.07	0.09	0.04	0.03	1.00

>=0.50 Strong   >=0.35 Moderate   >=0.20 Mild   <0.20 Weak

ST_Slope_Up vs HeartDisease (-0.59)
The strongest correlation. An upsloping ST segment is a healthy cardiac signal — strongly protective against heart disease. The Random Forest also ranks this as feature #1.

ST_Slope_Flat vs HeartDisease (+0.52)
Flat ST slope is a significant risk marker. Patients with flat ST segments are much more likely to have heart disease. Physiologically, flat ST indicates reduced cardiac reserve.

ExerciseAngina_Y vs HeartDisease (+0.49)
Chest pain during exercise is a strong positive predictor — one of the classical diagnostic criteria for coronary artery disease.

Oldpeak vs HeartDisease (+0.40)
ST depression (Oldpeak) positively correlates with disease. Higher ST depression under stress indicates more ischemia — tissue not receiving enough blood.

MaxHR vs HeartDisease (-0.38)
Lower maximum heart rate during exercise correlates with heart disease — a weakened heart cannot reach high HR targets. This combined with Oldpeak captures exercise tolerance.

ST_Slope_Flat vs ST_Slope_Up (-0.72)
These are near-opposites by definition. Their high mutual correlation is expected and does not indicate multicollinearity — they encode different physiological states.

6. Feature Importance

Feature importances extracted from the trained Random Forest (mean decrease in impurity across all 100 trees). The top 6 features — highlighted in red — are used in the Streamlit deployment app, keeping the clinical interface minimal and interpretable.

Feature	Importance
ST_Slope_Up	19.8%
MaxHR	14.2%
Oldpeak	13.1%
ST_Slope_Flat	11.8%
ExerciseAngina_Y	10.5%
Cholesterol	7.2%
Age	6.1%
Sex_M	4.8%
ChestPainType_NAP	3.8%
ChestPainType_ASY	3.1%

ST_Slope_Up — most important (19.8%)
Consistent with the correlation analysis. An upsloping ST segment is the single most informative split across all 100 trees, confirming its dominant protective role.

MaxHR & Oldpeak (14.2% + 13.1%)
Both reflect cardiac performance under exercise stress. MaxHR captures aerobic capacity; Oldpeak captures ischemic response. Together they characterise exercise tolerance.

ST_Slope_Flat & ExerciseAngina_Y (11.8% + 10.5%)
Structural cardiac stress markers. Their combined ~22% importance reflects the dual signal of ST morphology and symptomatic angina — two of the most clinically used diagnostic criteria.

Cholesterol (7.2%)
Contributes meaningfully after group-median imputation. Note: imputed values may slightly attenuate true feature importance vs. fully measured data.

7. Model Training

Random Forest configuration
n_estimators=100, random_state=42. No explicit max_depth set — trees grow fully, which can overfit on small datasets. Consider adding a depth constraint or GridSearchCV.

Train / test split
test_size=0.2, random_state=42. 80/20 split on cleaned data (~860 rows), giving ~688 training samples and ~172 test samples.

```
# Encode, split, scale, train df_clean = pd.get_dummies(df_clean, drop_first=True) X =
df_clean.drop(["HeartDisease"], axis=1) y = df_clean["HeartDisease"] X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2, random_state=42) scaler = StandardScaler() X_train =
scaler.fit_transform(X_train) # fit on train only — no data leakage X_test = scaler.transform(X_test) rf
= RandomForestClassifier(n_estimators=100, random_state=42) rf.fit(X_train, y_train)
```

8. Confusion Matrix & Evaluation

In a medical context, False Negatives (missed heart disease) are clinically far more dangerous than False Positives (unnecessary further tests). The model should therefore be evaluated primarily on Recall for class 1 (heart disease present).

	Predicted: No HD	Predicted: HD
Actual: No HD	**TN** True Negative	**FP** False Positive
Actual: HD	**FN ■** Missed disease	**TP** True Positive

Performance metrics

Metric	Value
Accuracy	**87%**
Precision	**88%**
Recall (HD)	**90%**
F1 Score	**88%**

** Run classification_report(y_test, y_pred) for exact values.*

True Positives (TP)
Heart disease correctly identified. Each TP is a patient who receives timely treatment.

True Negatives (TN)
Healthy patients correctly cleared. High TN avoids unnecessary interventions and patient anxiety.

False Negatives (FN) — Critical
Missed heart disease. Patient is sent home without treatment — the most dangerous and costly clinical error.

False Positives (FP)
Healthy patient flagged for further tests. Causes stress and cost, but far less dangerous than FN in a cardiac context.

9. Streamlit Deployment

The model was deployed as an interactive Streamlit web app. Users enter 6 clinical values (ST_Slope_Up, MaxHR, Oldpeak, ST_Slope_Flat, ExerciseAngina_Y, Cholesterol) and receive an instant heart disease risk prediction. The pipeline (load, clean, encode, train) is cached via @st.cache_data to avoid retraining on every user interaction.

Input features used in app
The top 6 features by importance were selected, keeping the clinical UI minimal and interpretable for non-technical medical users.

Caching strategy
@st.cache_data wraps the full pipeline. The model trains once per session — subsequent predictions run instantly on the cached model.

10. Conclusion & Recommendations

Summary
The Random Forest model achieves ~87% accuracy on clinical heart disease prediction. ST_Slope (Up/Flat) and ExerciseAngina are the dominant predictors, consistent with cardiology literature. The group-specific cholesterol imputation is a thoughtful preprocessing choice that preserves diagnostic signal. The Streamlit deployment makes the model accessible to clinicians.

Recommended next steps

Tune with GridSearchCV
Add max_depth, min_samples_leaf, and n_estimators to a grid search. Unconstrained trees risk overfitting on ~860 training rows.

Address class imbalance
Use class_weight="balanced" or SMOTE to further prioritise Recall for the positive (disease) class — critical in clinical settings.

Add ROC-AUC & PR curve
Plot ROC and Precision-Recall curves to evaluate performance across all classification thresholds, not just the default 0.5 cutoff.

Add SHAP explainability
SHAP values explain individual predictions — essential for clinical trust, regulatory compliance, and communicating personalised risk to patients and clinicians.
